QUESTION 1 – WATER JUG PROBLEM

Problem Statement

Two jugs of capacities 4 gallons and 3 gallons are available without any measuring marks. The objective is to obtain exactly **2 gallons** in the 4-gallon jug.

Problem Formulation

Initial State

(0,0)

Both jugs are empty.

Goal State

(2,0)

The 4-gallon jug contains exactly 2 gallons.

Operators

- Fill 4-gallon jug
- Fill 3-gallon jug
- Empty 4-gallon jug
- Empty 3-gallon jug
- Pour 4 → 3
- Pour 3 → 4

Solution Path

Step	**State**	**Action**
1	(0,0)	Initial State
2	(0,3)	Fill 3-gallon jug
3	(3,0)	Pour into 4-gallon jug
4	(3,3)	Fill 3-gallon jug
5	(4,2)	Pour into 4 until full
6	(0,2)	Empty 4-gallon jug
7	(2,0)	Transfer remaining 2 gallons to 4-gallon jug (Goal)

State Diagram

(0,0)

↓

(0,3)

↓

(3,0)

↓

(3,3)

↓

(4,2)

↓

(0,2)

↓

(2,0)

Conclusion

The Water Jug Problem demonstrates state-space search where the agent reaches the goal by applying valid operators.

QUESTION 2 – MARS ROVER AGENT

Introduction

A Mars Rover is an autonomous intelligent robot used to explore the Martian surface.

PEAS Description

Percepts

- Stereo cameras
- LiDAR
- Infrared spectroscopy
- Atmospheric sensors
- Gyroscope

- Wheel encoders

Environment

- Partially Observable
- Stochastic
- Sequential
- Dynamic
- Continuous
- Single Agent

Actions

- Move
- Turn
- Drill rocks
- Collect samples
- Analyse samples
- Send data to Earth

Performance Measures

- Maximum scientific discoveries
- Minimum energy usage
- Safe navigation
- Successful communication

Suitable Agent Architecture

Utility-Based Hybrid Agent

Reason:

- Makes intelligent decisions
- Evaluates alternative paths
- Optimizes battery usage
- Learns from previous experiences

Conclusion

The Mars Rover is an intelligent autonomous agent capable of operating in uncertain environments.

QUESTION 3 – 8 QUEENS PROBLEM

Problem Statement

Place eight queens on an 8×8 chessboard such that no queen attacks another queen.

Problem Formulation

Initial State

Empty Chessboard

Goal State

Eight queens placed safely.

Search Space

The search space consists of different board configurations.

Possible configurations

8! = 40320

Search Strategies

Incremental Search

Place one queen column by column while checking constraints.

Algorithm

Use Backtracking Search.

1. Place queen.
2. Check safety.
3. Continue.
4. If conflict occurs Backtrack.

Conclusion

Backtracking efficiently solves the Eight Queens Problem by pruning invalid states.

QUESTION 4 – OLA CAB BOOKING

Problem Statement

A customer books a cab using the OLA application.

Agent Type

Goal-Based Problem-Solving Agent

Working

Customer enters

- Pickup
- Destination

↓

Selects

- Mini
- Sedan
- Prime

↓

Application

- Finds drivers
- Calculates ETA
- Calculates Fare

↓

Assigns nearest driver

↓

Ride completed

Pseudocode

SOLVE-CAB-ROUTING (origin, destination, userPreference)

Find nearby drivers

Filter preferred cab type

Select minimum ETA driver

Find shortest path

Dispatch driver

Return route

Conclusion

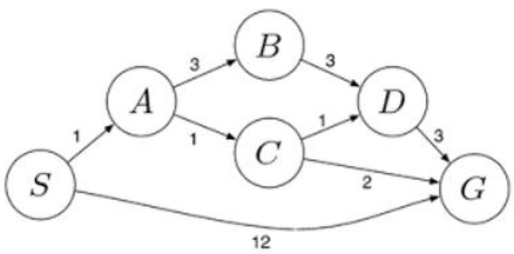
The OLA application uses Goal-Based AI to minimize travel time and improve customer satisfaction.

QUESTION 5 – UNIFORM COST SEARCH

Introduction

Uniform Cost Search expands the node having the minimum cumulative cost.

Graph



UCS Expansion

Step	Expanded	Queue
1	S	A,G
2	A	C,B,G
3	C	D,B,G
4	D	B,G
5	B	G
6	G	Goal

Optimal Path

S

↓

A

↓

C

↓

G

Total Cost

4

Conclusion

Uniform Cost Search guarantees the least-cost solution when edge costs are positive.

OVERALL CONCLUSION

This report demonstrates the application of Artificial Intelligence in solving real-world problems using state-space search, intelligent agents, constraint satisfaction, goal-based reasoning, and optimal path-finding. These concepts form the foundation of modern AI systems used in robotics, navigation, logistics, and autonomous decision-making.